LABORATORY REPORT

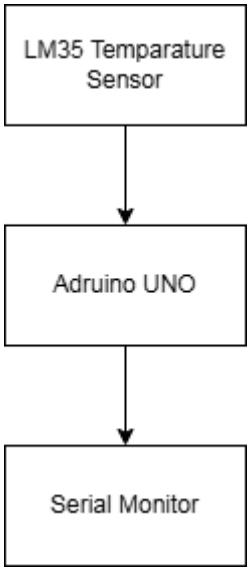
EXERCISE 1: TEMPERATURE SENSOR – LM35

Arduino & Embedded Systems Lab

GROUP 2		SE2045 – IOT102
Võ Nguyễn Thiên Phú	SE201591	Leader
Văng Khánh Khuyên	SE200481	Member
Nguyễn Văn Quốc Bảo	SE201682	Member
Võ Gia Huy	SE200956	Member

1. Block Diagram

The following block diagram illustrates the system architecture of the LM35 temperature sensor connected to an Arduino UNO microcontroller:



2. Operating Principle

The LM35 is a precision integrated-circuit temperature sensor whose output voltage is linearly proportional to the Celsius temperature, with a conversion factor of 10 mV/°C.

The LM35 sensor transmits a voltage signal corresponding to the ambient temperature to the analog pin of the Arduino UNO.

The Arduino reads the analog value from the sensor using the analogRead() function, producing an ADC value in the range of 0 to 1023.

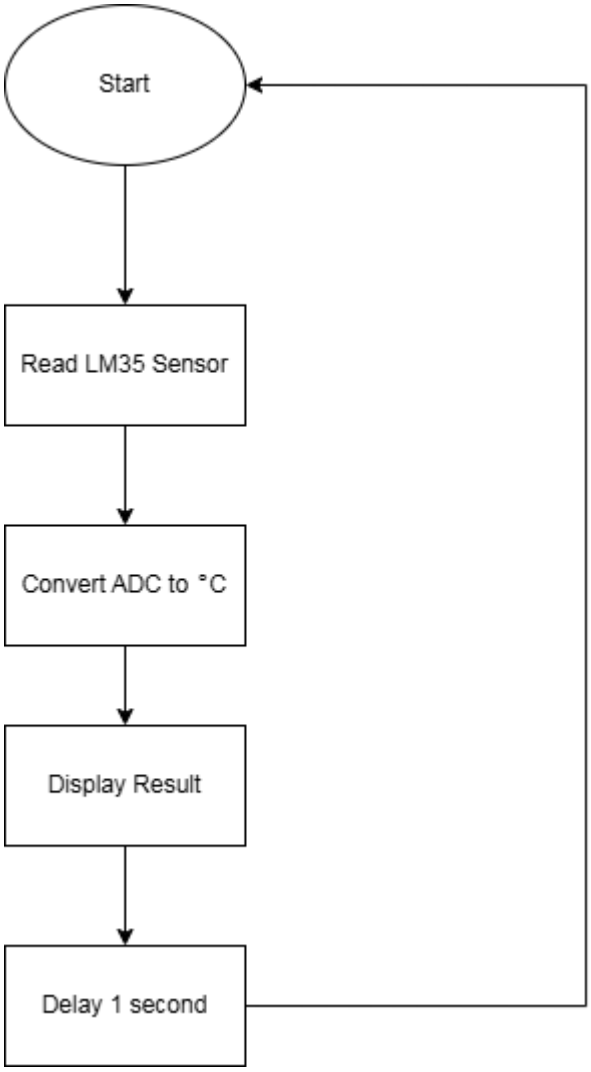
The Arduino then converts the ADC value into a temperature using the following formula:

```
Temperature (°C)  =  ADC × 500 / 1023
```

The computed temperature value is then displayed on the Serial Monitor for the user to observe in real time.

3. Program Flowchart

The flowchart below describes the logical sequence of operations performed by the Arduino firmware:



4. Source Code

The following C++ sketch is uploaded to the Arduino UNO to implement the temperature measurement system:

```
int temValue;
int tempPin = A1;

void setup() {
  Serial.begin(9600);
}

void loop() {
  temValue = analogRead(tempPin);
  float tempCelsius =
        temValue * 500.0 / 1023.0;
  Serial.print("Temperature: ");
  Serial.println(tempCelsius);
  delay(1000);
}
```

4.1 Code Explanation

- setup(): Initializes serial communication at a baud rate of 9600 bps.
- analogRead(): Reads the analog voltage signal from the LM35 sensor on pin A1.
- tempCelsius: Stores the converted temperature value in degrees Celsius.
- Serial.println(): Transmits and displays the result on the Serial Monitor.
- delay(): Pauses program execution for 1000 milliseconds (1 second) between readings.

5. Applications

The LM35-based temperature sensing system has a wide range of practical applications, including:

- Indoor and outdoor room temperature monitoring systems.
- Smart Home automation and environmental control.
- Temperature alarm and alert devices.
- Automatic fan speed control systems.
- IoT-based environmental monitoring projects.

Conclusion

This laboratory exercise provides students with a practical understanding of how to interface the LM35 temperature sensor with an Arduino UNO microcontroller to measure and monitor ambient temperature. Through this exercise, students gain hands-on experience with analog

signal reading, ADC-to-temperature conversion, and serial data output — fundamental skills in embedded systems and IoT development.